UNIVERSITY OF CAMBRIDGE

NST1
NATURAL SCIENCES TRIPOS Part IB

Friday 10 June 2022 14.00 to 17.00 (*3 hours plus any pre-agreed individual adjustment*)

# Paper PHA/WP

PHARMACOLOGY WRITTEN PRACTICAL PAPER

*Answer **all four** questions. Each **question** is worth **20 marks in total**.*

*Word process your answers in one of the following formats: .doc, .docx, .rtf or .txt (or with permission, hand write)*

*Extended Answer questions (2) and (4) have a **750 word** limit*

*Provide each main question answer in a single separate document, one for Q1, one for Q2, one for Q3, one for Q4, so four documents in total, named Candidate number_Question number, e.g. 7850X_Q1.docx*

*Where necessary to perform, **show your mathematical workings**. You may hand draw mathematical workings (or draw using a tablet), scan or photograph them, and clearly label them e.g. "Q1D mathematical working". The same applies to any figures to be drawn for the extended answer question. Each properly-labelled figure (maximum size 1GB) needs to be uploaded separately to Inspera using the upload tool, using a range of formats (e.g. .jpg, .png,.pdf).*

*You are not required to and should not plot any graphs. You will not be penalised for minor inaccuracies in reading values from graphs provided in questions.*

*Follow the University's guidance on Inspera regarding how to upload your documents*

*Calculator – students are permitted to use an approved calculator*

## Question 1

***NOTE: You will not be penalised for minor inaccuracies in estimating/reading values from graphs.***

A team of scientists is studying the pharmacology of γ-aminobutyric acid type-B ($GABA_B$) receptors in rat hippocampal brain slices. To begin with they perform a functional assay that measures a fluorescent signal generated in the neurones of the brain slice in response to changes in levels of second messenger. The scientists obtained dose response curves using the known $GABA_B$ agonist baclofen in the absence and presence of two different doses of a novel antagonist, Compound X (**Figure 1**).

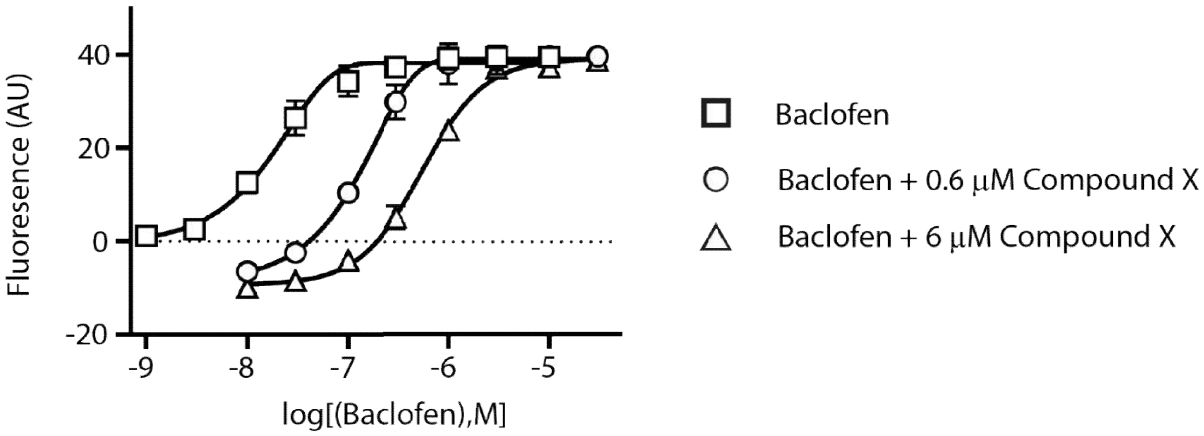


***Figure 1.*** *Agonist-induced dose response curves measured in terms of a fluorescence read-out reporter assay performed on rat hippocampal brain slices.*

Based only on the results in **Figure 1**, the scientists concluded that there is a background signal generated by endogenous levels of GABA within the tissue. They also concluded that Compound X is a competitive antagonist.

**A) Why do the scientists think endogenous GABA is present? (1 mark)**

**B) Provide an alternative receptor-based explanation for the background signal and suggest a way to test this hypothesis? (2 marks)**

Using the same assay additional dose response curves were obtained for baclofen at alternative concentrations of Compound X. The baclofen concentrations to achieve half-maximal responses in the presence of the various concentrations of Compound X were then used to establish dose ratios and create a Schild plot, as shown in **Figure 2**.

**[Question 1 continues on the NEXT PAGE]**

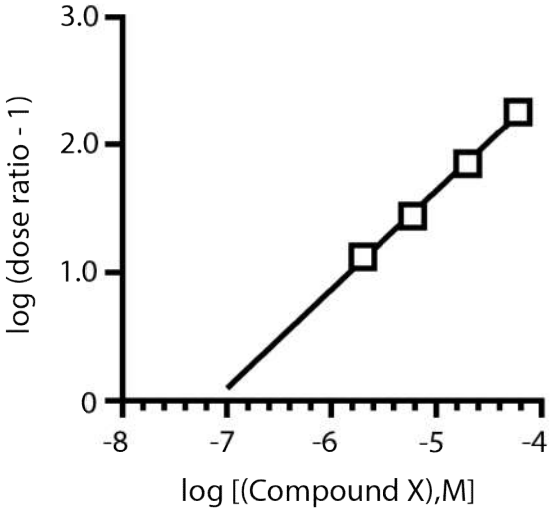


***Figure 2.*** *Relationship of baclofen dose ratios to different concentrations of Compound X.*

**C) Calculate the affinity constant of Compound X. (1 mark)**

Compound X contains an aryldiazirine functional group. For any molecules of Compound X that are bound to its receptor pocket, the aryldiazirine reactive group forms a direct irreversible covalent link to the receptor with an efficiency of 50% upon a very brief millisecond flashing with ultraviolet (UV) light at a wavelength of 340 nm. The scientists performed the following experiment on a hippocampal brain slice. First, they measured a baclofen dose response curve. Second, they incubated the hippocampal brain slice with Compound X at its $K_D$ concentration and applied the very brief UV flash. Third, after thoroughly flushing out all non-covalently bound Compound X, they measured another dose response curve to baclofen. The results of this experiment are shown in **Figure 3**.

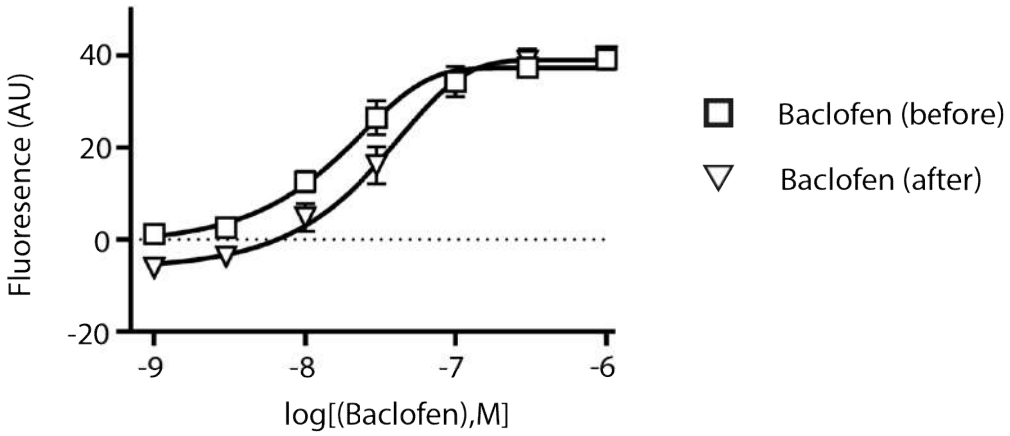


***Figure 3.*** *Agonist-induced dose response curves measured in terms of a fluorescence read-out reporter assay performed on rat hippocampal brain slices, before and after the slice was briefly flashed with UV light at 340nm wavelength in the presence of Compound X.*

**[Question 1 continues on the NEXT PAGE]**

**D) The scientists deduced that baclofen is a full agonist because the maximal response did not go down after the brain slice was treated with Compound X. Why can the scientists reach this conclusion from the data in Figure 3, yet they could not from the data in Figure 1? (1 mark)**

**E) Hypothetically, if baclofen were a partial agonist, would you expect the maximum response to drop by 25 % in Figure 3? Explain your answer (2 marks)**

To better understand the properties of the $GABA_B$ receptors in the hippocampal brain slices, the scientists decided to characterise the binding properties of baclofen and GABA to these receptors. To look only at cell surface receptors, they isolated/purified the surface membranes of the neurones in the hippocampal brain slice, and prepared these as suspensions with different doses of the radioligand [$^3$H]-baclofen. Membranes were washed using a filtration system to remove excess radioligand. A non-specific binding control was included. A specific binding curve is plotted in **Figure 4a**. Next, competition experiments were performed using 100 nM [$^3$H]-baclofen and increasing concentrations of GABA (**Figure 4b**).

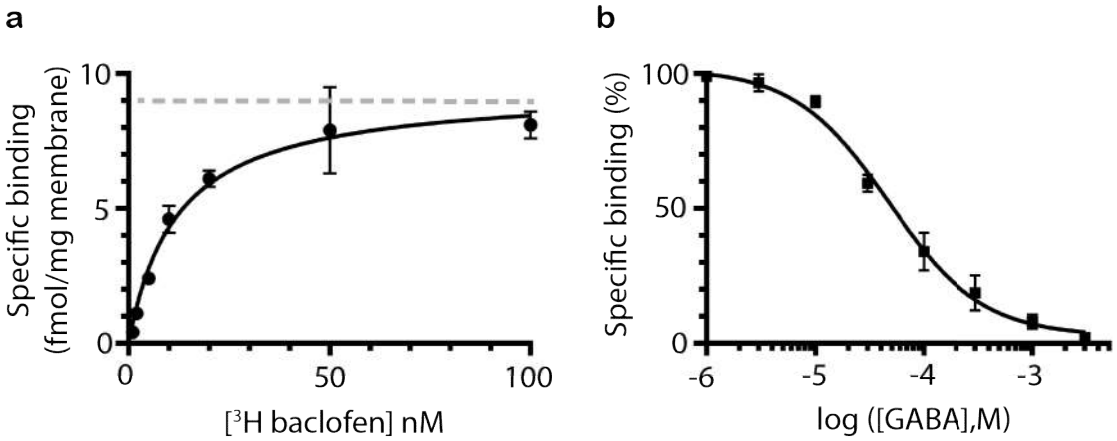


***Figure 4. a,*** *Radioligand binding curve for [$^3$H]-baclofen. Binding was measured in terms of femtomol (fmol) of radioligand binding per mg of surface membrane (fmol/mg membrane). Maximum specific binding is indicated by the dashed grey line.* ***b,*** *Competition radioligand binding experiment was performed with 100 nM of [$^3$H]-baclofen and the extent of specific binding was measured at increasing concentrations of GABA.*

**F) Without drawing any further graphs but using data presented in Figure 4, calculate the $K_A$ for GABA for $GABA_B$ receptors. (4 marks)**

In **Figure 4a** binding is presented in terms of femtomole (fmol) of radioligand binding per mg of cell surface membrane. In these experiments cell surface membrane accounts for 3.2 % of the original wet weight of the brain slices.

**G) How many receptors are there in each gram of brain slice tissue? Avogadro's number is $6.022 \times 10^{23}$ $mol^{-1}$. (2 marks)**

**[Question 1 continues on the NEXT PAGE]**

The scientists performed again the same [$^3$H]-baclofen labelling experiment as in **Figure 4a**, however this time they first pre-treated the tissue. To do this, before preparing the membranes they first incubated the intact hippocampal brain slice with Compound X at its $K_D$ concentration and applied a very brief UV flash, and then washed the intact tissue. Then they prepared the membranes and obtained the specific binding curve for [$^3$H]-baclofen. This time the maximal binding achieved was 7.1 fmol/mg.

**H) What proportion of receptors (as a percentage of the total) are occupied by Compound X when it is applied at its $K_D$ to the hippocampal brain slice, <u>before</u> the UV flash is applied? (2 marks)**

**I) What concentration of GABA is present in the tissue? (4 marks)**

**J) Why is the answer to (I) unlikely to be the real concentration of GABA in the hippocampal brain slice? (1 mark)**

**End of Question 1**

**2. Compare the different traits of drug action through ion channels versus G-protein-coupled receptors. (20 marks)**

**[Question 3 starts on the NEXT PAGE]**

## Question 3

Cambridge HT are testing the pharmacokinetics of a new anti-hypertensive drug, HT-003. HT-003 is rapidly absorbed after oral administration, and it has an oral bioavailability is 60%. To further understand the pharmacokinetic properties of HT-003, a healthy adult volunteer and a patient with renal insufficiency were each given a single 20 mg oral dose. **Figure 1** shows the plasma concentrations as a function of time after administration of HT-003. The plasma concentrations are also listed in **Table 1**.

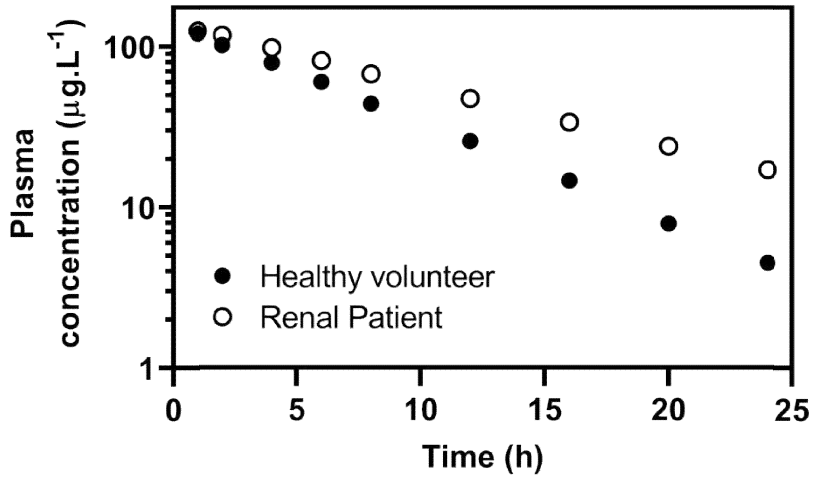


**Figure 1.** Plasma concentration versus time profile of HT-003 after a 20 mg dose in a healthy adult volunteer and a renal patient. The plasma concentrations are plotted on a logarithm scale. Data from healthy volunteer and renal patient are shown in filled and open circles, respectively.

**Table 1.** Plasma concentrations of HT-003 measured at time intervals after a single 20 mg dose administered to a normal volunteer and a renal patient.

| **Time (hours)** | **Plasma concentration ($\mu g.L^{-1}$)** | |
|---|---|---|
| | Healthy volunteer | Renal patient |
| **1** | 120 | 125.2 |
| **2** | 101 | 115 |
| **4** | 79.1 | 96.2 |
| **6** | 60.2 | 81.9 |
| **8** | 44.0 | 67.6 |
| **12** | 25.8 | 47.5 |
| **16** | 14.6 | 33.8 |
| **20** | 7.9 | 23.9 |
| **24** | 4.5 | 17.1 |

**A) Estimate the rate constant for elimination and the clearance for the healthy volunteer and for the renal patient from the data provided. (6 marks)**

**[Question 3 continues on the NEXT PAGE]**

During the course of the study, the urine voided between 12 and 16 hours was collected and analysed. The results showed that the normal volunteer produced 300 ml of urine that contained 0.97 $\mu g.ml^{-1}$ of HT-003, whereas the renal patient produced 218 ml of urine that contained 1.1$\mu g.ml^{-1}$ of HT-003.

**B) What can you conclude about the handling of HT-003 by the two individuals? (5 marks)**

Subsequently, HT-003 is given to the normal volunteer as a 10 mg oral dose every 8 hours.

**C) Calculate the maximum and minimum plasma concentrations of HT-003 at steady state for the normal individuals. (4 marks)**

**D) How should the dose of HT-003 be modified to achieve the same steady state maximum plasma concentration in the patient with renal insufficiency? (3 marks)**

**E) Despite rapid absorption of HT-003, the bioavailability was only 60% - explain. Can you suggest any alternative but non-parenteral route of administartion of HT-003 drug that could improve the bioavailability? (2 marks)**

**End of Question 3**

**4. Discuss with examples how the pharmacokinetic properties of drugs need to be optimised during their development. (20 marks)**

**END OF PAPER**